MVA RecVis'24: "CoVR-2: Automatic Data Construction for Composed Video Retrieval"

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Motivation

Gain of popularity of Composed Image Retrieval (CoIR) task to search for relevant images in huge dataset

- Librarians: research and cataloging
- Detectives: criminal investigation
- Content creators: create rich visual
- Travel planning: comprehensive destination visuals
- etc...

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Motivation

Gain of popularity of Composed Image Retrieval (CoIR) task to search for relevant images in huge dataset

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data privacy? inappropriate content? copyright?

Motivation

Most **CoIR** task require manually annotated datasets, but this limits scalability because

- Time-consuming
- Expensive

CoVR-2: Automatic Data Construction for Composed Video Retrieval:

- Scalable, automatic dataset creation methodology.
- Generate triplets from video-caption pairs, expanding the scope to include Composed Video Retrieval (CoVR)
- Build a novel model to respond to the CoVR task

CoVR-2

Method Overview of WebVid-CoVR

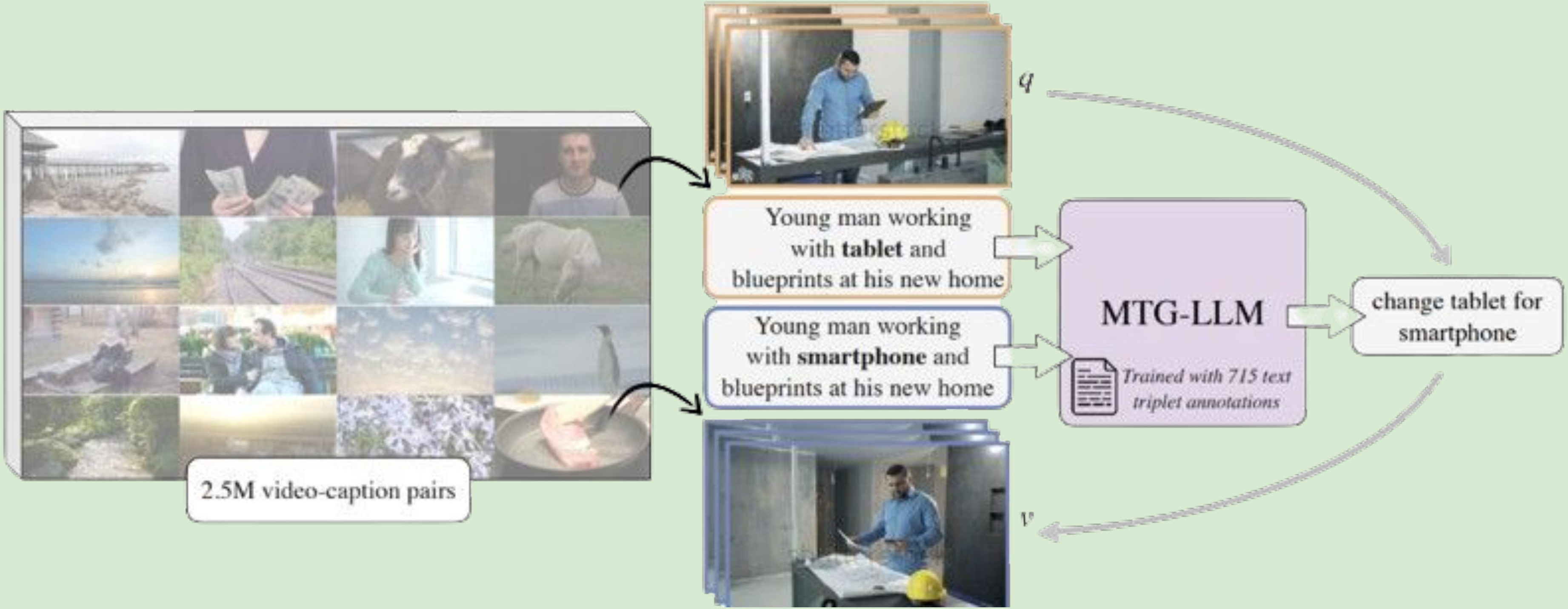

Fig. Method overview from [1]

CoVR-2

Created WebVid-CoVR-Test, a new CoVR benchmark that is manually annotated

Dataset	Type	#Triplets	#Unique visuals	#Unique words	Avg. text length	Domain
CIRR [8]		36,554	21,185	7,129	59.51	Natural
FashionIQ [9]		30,132	7,988	4,425	27.13	Fashion
CIRCO-Test [4]		800	-	870	50.94	Natural
LaSCo [7]		389,305	121,479	13,488	30.70	Natural
SynthTriplets18M [6]		18,000,000	-	-	-	Synthetic
CC-CoIR		3,315,773	356,582	28,183	24.65	Natural
WebVid-CoVR		1,644,276	130,559	25,654	23.36	Natural
WebVid-CoVR-Test		2,556	4,886	1,910	21.97	Natural

Fig. CoIR and CoVR Benchmark from [1]

CoVR-2

CoVR-BLIP-2 Architecture

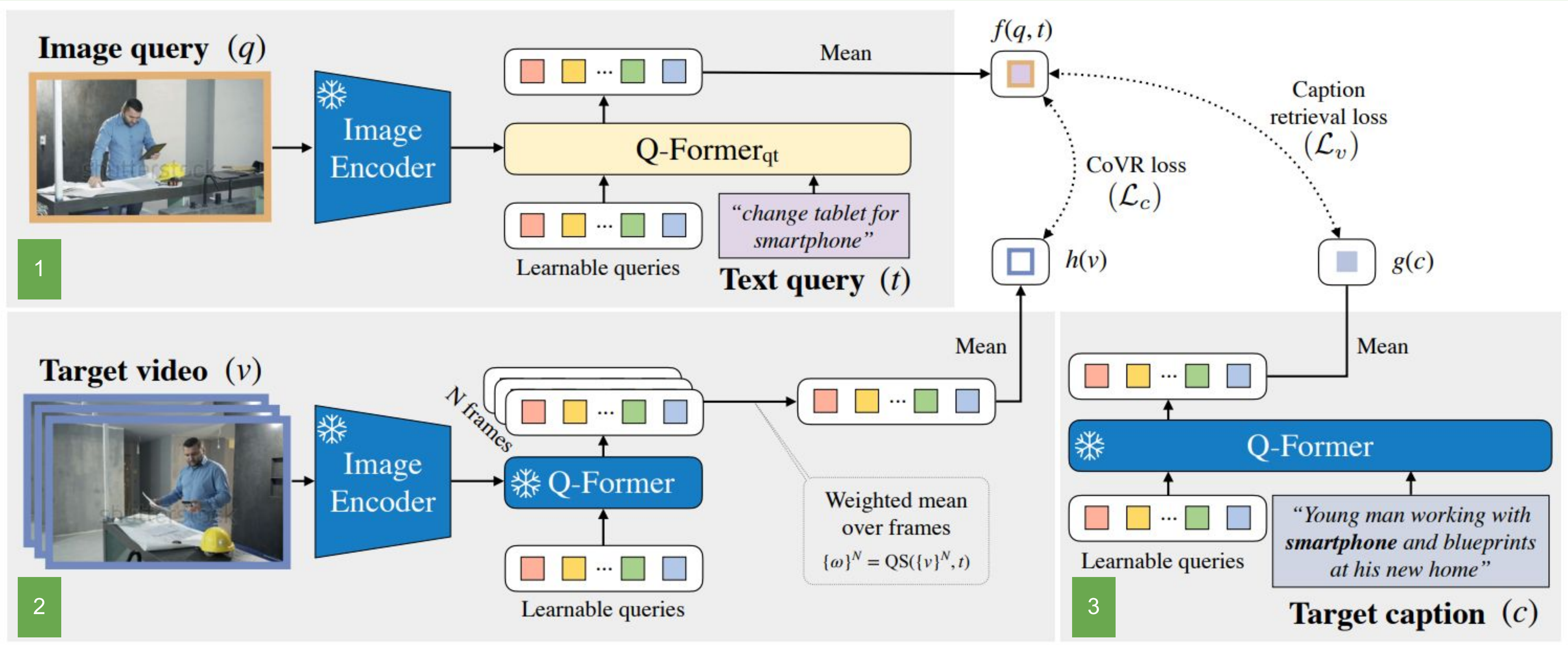

Fig. Model architecture of CoVR-BLIP-2 from [1]

CoVR-2

Showed good transfers results for CoIR (SoTA on CIRR and CIRCO in zero-shot)

Mode	Method	Year	Pretraining Data	CIRR Recall@K K=1	K=5	K=10	K=50	R_{subset}@K K=1	K=2	K=3
Zero Shot	GRB [23]	2023	-	24.19	52.07	65.48	85.28	60.17	79.13	89.34
	GRB+LCR [23]	2023	-	30.92	56.99	68.58	85.28	66.67	78.68	82.60
	TFCIR [23]	2023	-	32.82	61.13	71.76	85.28	66.63	78.58	82.68
	CASE [7]	2024	LaSCo	30.89	60.75	73.88	92.84	60.17	80.17	90.41
	CASE [7]	2024	LaSCo+COCO	35.40	65.78	78.53	94.63	64.29	82.66	91.61
	CIReVL [25]	2024	-	34.65	64.29	75.06	91.66	67.95	84.87	93.21
	LinCIR (ViT-L) [24]	2024	-	25.04	53.25	66.68	-	57.11	77.37	88.89
	LinCIR (ViT-G) [24]	2024	-	35.25	64.72	76.05	-	63.35	82.22	91.98
	CoVR-BLIP [58]	2024	WebVid-CoVR	38.48	66.70	77.25	91.47	69.28	83.76	91.11
	CoVR-BLIP-2	2024	CC-CoIR	43.35	**73.78**	83.66	96.07	**75.25**	**88.89**	**95.23**
	CoVR-BLIP-2	2024	WV-CC-CoVIR	**43.74**	73.61	**83.95**	**96.10**	72.84	87.52	94.39

Fig. State-of-the-art comparison on the CIRR on test set from [1]

CoVR-2

Showed good transfers results for CoIR (SoTA on CIRR and CIRCO in zero-shot)

Method	Year	Pretraining Data	mAP@K K=5	K=10	K=25	K=50
Random		-	0.00	0.00	0.00	0.00
SEARLE-XL [4]	2023	CC3M	11.68	12.73	14.33	15.12
CompoDiff (ViT-G) [6]	2023	ST18M	15.33	17.71	19.45	21.01
GRB+LCR [23]	2023	-	25.38	26.93	29.82	30.74
TFCIR [23]	2023	-	26.52	28.25	31.23	31.99
CIReVL [25]	2024	-	26.77	27.59	29.96	31.03
LinCIR (ViT-G) [24]	2024	-	19.71	21.01	23.13	24.18
CoVR-BLIP [58]	2024	WebVid-CoVR	21.43	22.33	24.47	25.48
CoVR-BLIP-2	2024	CC-CoIR	23.50	24.66	27.05	28.04
CoVR-BLIP-2	2024	WV-CC-CoVIR	**28.29**	**29.55**	**32.18**	**33.26**

Fig. State-of-the-art comparison on the CIRCO test set from [1]

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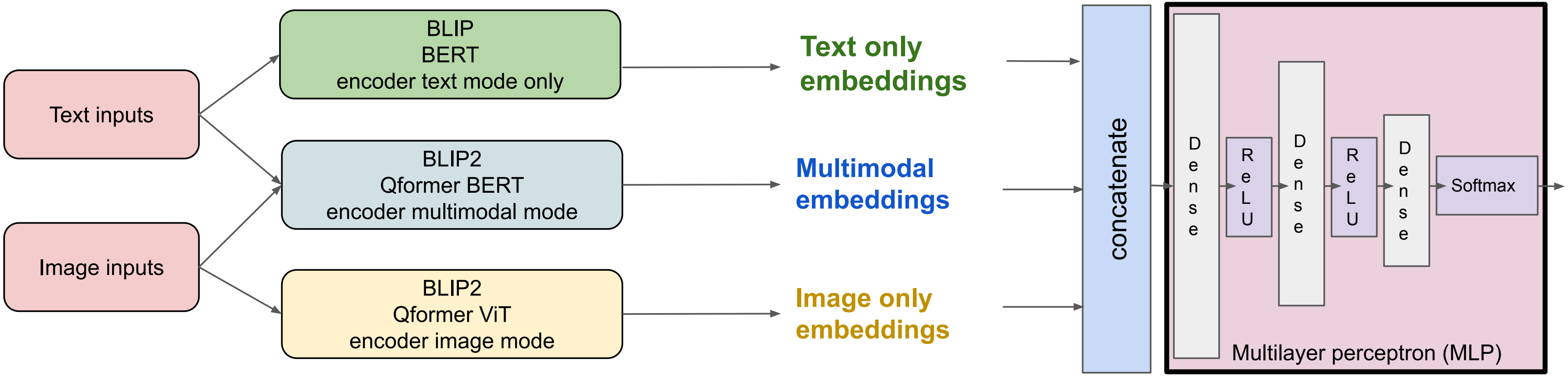
Works

- Develop a MLP to balance contributions of image, text, and multimodal embeddings

(MLP) Weighted attention framework: *w1* * Visual embeddings + *w2* * Textual embeddings + *w3* * Hybrid embeddings



Works

- Add a consistency loss function to align multimodal, image, and text embeddings in the embedding space

$$\mathcal{L}_v(\mathcal{B}) = -\sum_{i\in\mathcal{B}} \log\left(\frac{e^{S_{i,i}/\tau}}{\alpha\cdot e^{S_{i,i}/\tau} + \sum_{j\neq i} e^{S_{i,j}/\tau} w_{i,j}}\right) - \sum_{i\in\mathcal{B}} \log\left(\frac{e^{S_{i,i}/\tau}}{\alpha\cdot e^{S_{i,i}/\tau} + \sum_{j\neq i} e^{S_{j,i}/\tau} w_{j,i}}\right)$$

- Consistency loss integration with 3 loss components :
  - **si-ti loss:** Align query (source image) with target image features
  - **si-text loss:** Align query with text features.
  - **si-img loss:** Align query with image features

si-img loss + si-text loss + si-ti loss

Results

I/ Reproduction of the results

Mode	Method	Year	Pretraining Data	CIRR Recall@K K=1	Recall@K K=5	Recall@K K=10	Recall@K K=50	R_{subset}@K K=1	R_{subset}@K K=2	R_{subset}@K K=3
	CoVR-BLIP-2	2024	-	**50.87**	80.80	88.84	98.00	76.70	90.31	95.45
	CoVR-BLIP-2	2024	CC-CoIR	50.63	81.04	**89.35**	**98.15**	76.53	**90.43**	**96.00**

model: blip2-l-coco

Recall@	Score (Accuracy Perc.)
1	30.241
2	42.506
5	59.735
10	71.807
50	90.892

model: blip2-l-coco_coir

Recall@	Score (Accuracy Perc.)
1	34.337
2	47.253
5	63.928
10	74.554
50	92.458

baseline

Configurations (GCP)

- *1 NVIDIA V100 GPU*
- *76 RAM CPU*
- *400GB Disk Memory*

Configurations (Code)

- *batch size = 32*
- *epoch = 2*
- *others are same as original*

Results

II/ Balancing contributions of image, text, and multimodal embeddings with MLP

1/ Equal contributions

1/3 * Visual embeddings + *1/3* * Textual embeddings + *1/3* * Multimodal embeddings

Recall@	Score (Accuracy Perc.)
1	23.976
2	34.627
5	51.518
10	64.169
50	87.639

Recall@	Score (Accuracy Perc.)
1	34.337
2	47.253
5	63.928
10	74.554
50	92.458

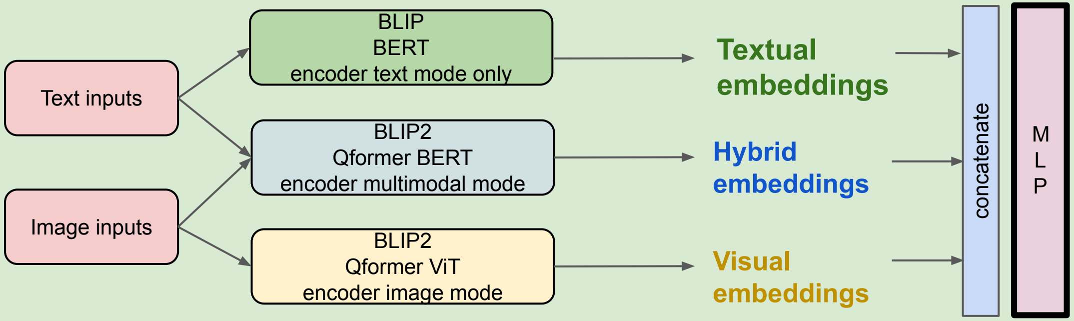
baseline

Results

II/ Balancing contributions of image, text, and multimodal embeddings with MLP

2/ MLP contributions balancing

(MLP) framework: *w1* * Visual embeddings + *w2* * Textual embeddings + *w3* * Multimodal embeddings




```
success All tests passed.
```

Recall@	Score (Accuracy Pe
1	33.566
2	47.084
5	63.518
10	74.867
50	92.723

Recall@	Score (Accuracy Perc.)
1	34.337
2	47.253
5	63.928
10	74.554
50	92.458

baseline

Results

II/ Balancing contributions of image, text, and multimodal embeddings with MLP

2/ MLP contributions balancing

(MLP) framework: *w1* * Visual embeddings + *w2* * Textual embeddings + *w3* * Multimodal embeddings

```
success All tests passed.
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Recall@	Score (Accuracy Pe
1	33.566
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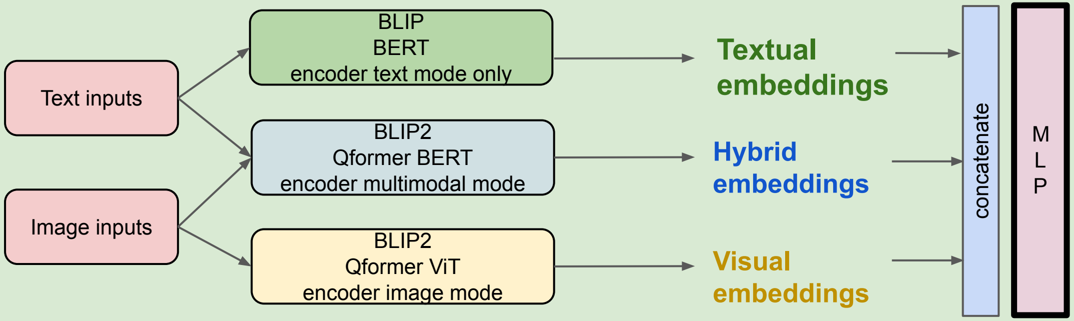

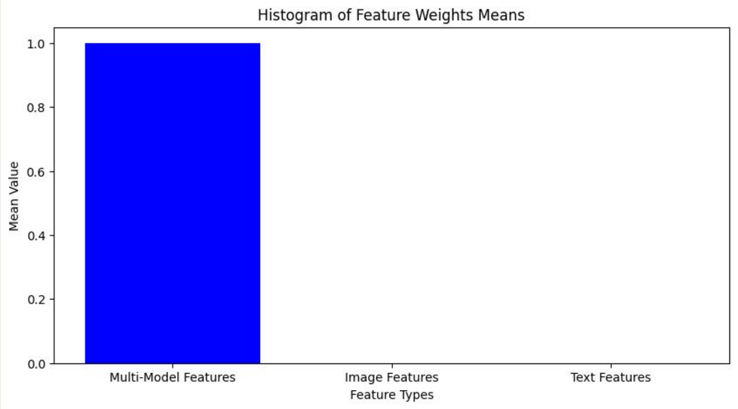

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Results

III/ Consistency loss

1/ Equi-balancing

- Combine features using learned weights:

w1 * Visual embeddings + *w2* * Textual embeddings + *w3* * Multimodal embeddings

- Consistency loss integration with 3 loss components :
  - **si-ti loss:** Align query (source image) with target image features
  - **si-text loss:** Align query with text features.
  - **si-img loss:** Align query with image features

(si-img loss + si-text loss + si-ti loss)/3

```
success All tests passed.
```

Recall@	Score (Accuracy Perc.)
1	26.337
2	37.711
5	54.313
10	66.386
50	89.47

Recall@	Score (Accuracy Perc.)
1	34.337
2	47.253
5	63.928
10	74.554
50	92.458

baseline

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Results

III/ Consistency loss

2/ Fixed loss balancing

- Combine features using learned weights:

w1 * Visual embeddings + *w2* * Textual embeddings + *w3* * Multimodal embeddings

- Consistency loss integration with 3 loss components :
    - **si-ti loss:** Align query (source image) with target image features
    - **si-text loss:** Align query with text features.
    - **si-img loss:** Align query with image features

1/4 * si-img loss + *1/4* * si-text loss + *1/2* * si-ti loss

```
success All tests passed.
```

Recall@	Score (Accuracy Perc.)
1	27.036
2	38.771
5	55.012
10	66.867
50	89.181

Recall@	Score (Accuracy Perc.)
1	34.337
2	47.253
5	63.928
10	74.554
50	92.458

baseline

Conclusion

- **Next steps:**
  - ➢ Work on the improvement of captions as the paper pointed out (Section 3.2) by using Capfilt introduced in BLIP
  - ➢ Consider more diverse weighting configurations for MLP
  - ➢ Use other consistency loss functions
  - ➢ Experiment with more epoch and with greater batch size

- **Issues:**
  - ➢ Significant time was devoted to ensuring a functional and stable environment.
  - ➢ Computational running time remains a major bottleneck.
  - ➢ Need bigger CPUs (or more GPUs) to improve our results which is a version adapted to our hardware configurations

References

[1] L. Ventura, A. Yang, C. Schmid, and G. Varol, “CoVR-2: Automatic Data Construction for Composed Video Retrieval”, AAAI, 2024. arXiv preprint arXiv:2308.14746.

[2] L. Ventura, A. Yang, C. Schmid, and G. Varol, “Covr: Learning composed video retrieval from web video captions”, AAAI, 2023. arXiv preprint arXiv:2308.14746.

[3] J. Li, D. Li, C. Xiong, and S. Hoi, “Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation,” 2022. arXiv preprint arXiv:2201.12086.

[4] J. Li, D. Li, S. Savarese, and S. Hoi, “Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” 2023. arXiv preprint arXiv:2301.12597.

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APPENDIX

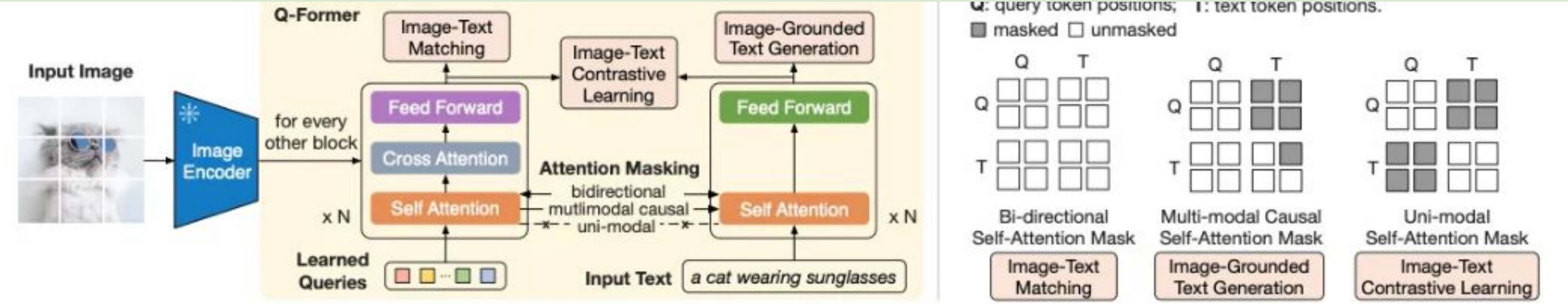

Figure 2. (**Left**) Model architecture of Q-Former and BLIP-2's first-stage vision-language representation learning objectives. We jointly optimize three objectives which enforce the queries (a set of learnable embeddings) to extract visual representation most relevant to the text. (**Right**) The self-attention masking strategy for each objective to control query-text interaction.

Q-Former